

BAYESIAN ANALYSIS

Homework optional

Arnau García

March 8, 2024

Statements: We observe the number of times that an event occurs. The prior information about the expected value of the counts of this event that we have is that the expected value will be between 4 and 6 with high probability. After observing a sample of size 5, $x = 1, 2, 2, 8, 10$, you have to decide whether this data have been generated by a Poisson model or by a Geometric model using a bayesian hypothesis test. And then, using the Bayesian model averaging approach, plot the posterior predictive distribution for a new future value and compute a point estimate and a 95% credible interval for the prediction. Explain all the steps you take in detail.

Useful information

If X is a random variable which follows a Poisson distribution with parameter λ , $\text{Poisson}(\lambda)$, then its probability function is:

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

with $E(X) = \lambda$, $V(X) = \lambda$, $x \in \{0, 1, 2, \dots\}$ and $\lambda \in R_+$.

If X is a random variable which follows a Geometric distribution with parameter θ , $\text{Geom}(\theta)$, then its probability function is:

$$p(x) = \theta(1 - \theta)^x$$

with $E(X) = \frac{1-\theta}{\theta}$, $V(X) = \frac{1-\theta}{\theta^2}$, $x \in \{0, 1, 2, \dots\}$ and $\theta \in [0, 1]$.

SOLUTION:

The very first step is to build the two Bayesian models. Let us start developing the Bayesian model corresponding to the Poisson model, from now on this model will be the first model, denoted by M_1 .

- *Bayesian model and further analysis for the Poisson model:*

Clearly, the statistical model in this case is given by the Poisson distribution and is

$$p_1(x|\lambda) = \frac{e^{-\lambda} \lambda^x}{x!}.$$

Now, our goal is to choose the prior distribution. Our prior information is that the expected value of the counts will be between 4 and 6 with a high probability. Since the expected value of a Poisson distribution is exactly $E(X) = \lambda$, we can choose the prior distribution for model directly λ . Notice that $\lambda \in (0, +\infty)$, thus, a good idea is work with a Gamma distribution. Besides of having the desired support, the Gamma distribution is a conjugate distribution of the Poisson, which is a useful property that we will use later on. Having decided to use a $Gamma(\alpha, \beta)$ distribution (with the rate parametrization), the next step is choosing the parameters α, β . Notice that, if $Z \sim Gamma(\alpha, \beta)$, then $E(Z) = \alpha/\beta$ and $Var(Z) = \alpha/\beta^2$. Since we want a prior distribution such that $E(X) = \lambda$ is between 4 and 6 with a high probability, we find parameters α, β such that

$$\begin{cases} E(Z) = \frac{\alpha}{\beta} = 5 \\ \sqrt{Var(Z)} = \frac{\sqrt{\alpha}}{\beta} = \frac{1}{2}. \end{cases}$$

It is, a prior such that λ is centered in 5 and such that the standard deviation is $1/2$, with this standard deviation we put approximately the 95% of the probability between 4 and 6. Solving the previous system of equations we obtain $(\alpha = 100, \beta = 20)$. Thus, the prior distribution of the Poisson model is $\pi(\lambda) \sim Gamma(100, 20)$. And then, we have the Bayesian model for the Poisson model built. Moreover, using that we have chosen a conjugate prior we know how is the posterior distribution, which is a $Gamma(100 + \sum_{i=1}^n x_i, 20 + n)$.

Let us now compute the likelihood function in this case:

$$L_1(\lambda|x) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-5\lambda} \lambda^{23}}{4 \cdot 8! \cdot 10!},$$

where we have used that $n = 5, \sum_{i=1}^5 x_i = 23$. Let us now summarize what we have for this model in Table [1](#).

Model 1	Likelihood	Prior	Posterior
$Poisson(\lambda)$	$\frac{e^{-5\lambda}\lambda^{23}}{4 \cdot 8! \cdot 10!}$	$Gamma(100, 20)$	$Gamma(123, 25)$

Table 1: Table with the information referring to the Poisson model.

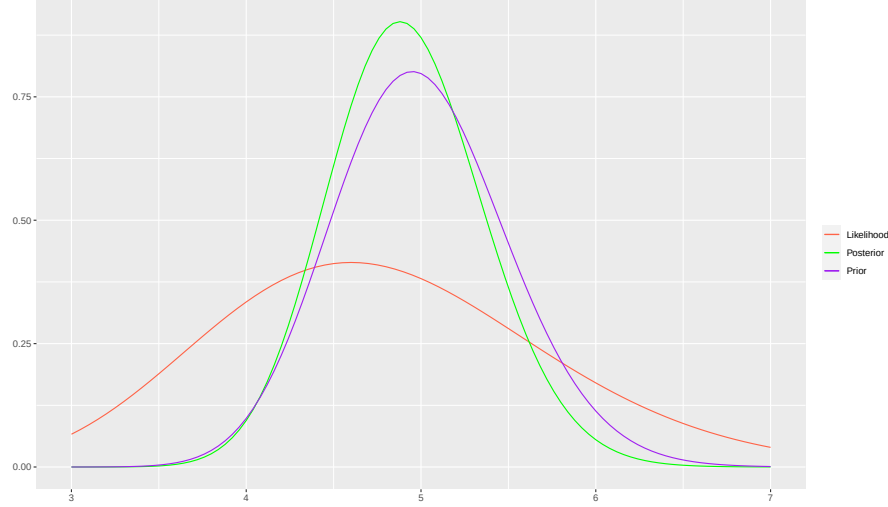


Figure 1: The likelihood function together with the prior and posterior distribution, for the Poisson model.

The standardized likelihood is:

$$L_{std1}(\lambda|x) = \frac{e^{-5\lambda}\lambda^{23}}{4 \cdot 8! \cdot 10!} \cdot \frac{11920928955078125}{8834477652} = \frac{95367431640625e^{-5\lambda}\lambda^{23}}{41363226782215962624}.$$

With the quantity above and with the prior and posterior distributions one can develop a plot collecting those three key components in order to have an intuition on what is happening in this Bayesian model. See the plot in Figure 1. As we were expecting, the posterior is a compromise between the prior and the likelihood. Moreover, in this case the prior is more informative than the likelihood, and then the posterior is more similar to the prior than to the likelihood. Notwithstanding, the model seems to make sense.

- *Bayesian model and further analysis for the Geometric model:*

Now we will be dealing with the Geometric model, which is the second model and will be denoted as M_2 from now on. The statistical model in this case is given by

$$p(x|\theta) = \theta(1 - \theta)^x.$$

Where $\theta \in [0, 1]$ and $E(X) = (1 - \theta)/\theta$, $Var(X) = (1 - \theta)/\theta^2$. Now, we choose the prior distribution. In this case the space of the parameter is $[0, 1]$, which suggests us that the Beta distribution could be a good choice. Moreover, the Beta is a conjugate prior in this case. Notice now that we have to be careful with the prior information. Our prior information is that $E(X) = (1 - \theta)/\theta$ is between 4 and 6 with a high probability. Hence, we have to transform this information in terms of the parameter θ , which is the parameter modeled by the prior distribution. Notice that $(1 - \theta)/\theta$ is a continuous function of θ in $[4, 6]$. So, let us transform the information in terms of θ :

$$\left\{ \begin{array}{l} \frac{1-\theta}{\theta} = 4 \\ \frac{1-\theta}{\theta} = 6 \end{array} \right\} \iff \left\{ \begin{array}{l} \theta = \frac{1}{5} \\ \theta = \frac{1}{7} \end{array} \right.$$

And then, the prior information is equivalent to say that θ is between $1/7$ and $1/5$ with a high probability. Thus, let us choose the proper parameters a, b of $Beta(a, b)$. Notice that if $Z \sim Beta(a, b)$, then $E(X) = a/(a + b)$, $Var(X) = (ab)/(a + b)^2(a + b + 1)$. With the aim of choosing a, b , we use the following system of equations:

$$\left\{ \begin{array}{l} E(Z) = \frac{a}{a+b} = \frac{6}{35} \\ \sqrt{Var(Z)} = \sqrt{\frac{ab}{(a+b)^2(a+b+1)}} = \frac{3}{200}. \end{array} \right.$$

We have proposed the above system to find a, b such that the prior distribution is centered in the midpoint of the interval $[1/7, 1/5]$ and has a standard deviation of $3/200$. Solving the system we obtain $a = 185306/1715$, $b = 2686937/5145$. With these parameters, we have approximately the 95% of the probability between $1/7$ and $1/5$. Thus, the prior distribution is $Beta(185306/1715, 2686937/5145)$. Using that we are dealing with a conjugate prior, we know in advance the posterior distribution which is $Beta(a + n, b + \sum_{i=1}^n x_i)$.

Let us compute the likelihood in this case:

$$L_2(\theta|x) = \prod_{i=1}^n \theta(1 - \theta)^{x_i} = \theta^n (1 - \theta)^{\sum_{i=1}^n x_i} = \theta^5 (1 - \theta)^{23}$$

As we did it before, we summarize all the key quantities in a table, see in Table [2](#).

Model 2	Likelihood	Prior	Posterior
$Geom(\theta)$	$\theta^5(1 - \theta)^{23}$	$Beta(185306/1715, 2686937/5145)$	$Beta(193881/1715, 2805272/5145)$

Table 2: Table with the information referring to the Geometric model.

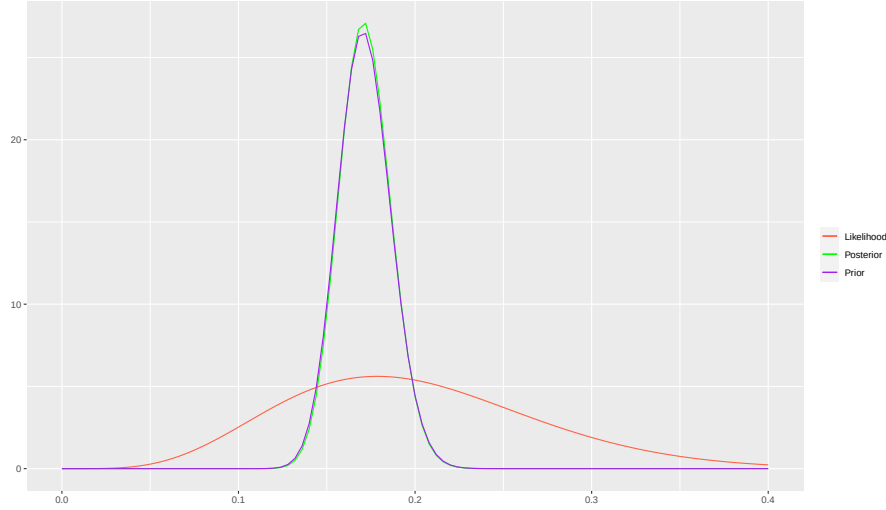


Figure 2: The likelihood function together with the prior and posterior distribution, for the Geometric model.

In this case the standardized likelihood is as follows:

$$L_{std_2}(\theta|x) = 2850120 \cdot \theta^5(1 - \theta)^{23}.$$

We proceed as above, drawing the standardized likelihood function, and the prior and posterior distribution together. The reader can see the graphic in the Figure 2. In this case the prior is much more informative than the likelihood. As we can see in the Figure 2, the posterior distribution (green line) overlaps with the prior distribution (purple line).

- *Hypothesis testing:*

Once we have presented the two Bayesian models, now our objective is to expose the hypothesis test and develop this test to decide whether the data have been generated by a Poisson or a Geometric model. We have the following hypothesis test:

$$\begin{cases} H_1 : & M_1 = \{p_1(\tilde{x}|\lambda), \lambda \in (0, +\infty)\}, \pi_1(\lambda) \sim \text{Gamma}(100, 20), \quad P(H_1) = P(M_1) = \frac{1}{2} \\ H_2 : & M_2 = \{p_2(\tilde{x}|\theta), \theta \in [0, 1]\}, \pi_1(\theta) \sim \text{Beta}(a, b), \quad P(H_2) = P(M_2) = \frac{1}{2} \end{cases} \quad (1)$$

Where $a = 185306/1715$, $b = 2686937/5145$ as we said before. The reader can observe that we have putted the same probability for each model (i.e. for each hypothesis). The next step is to calculate the posterior probability for each model:

$$P(M_i|x) = \frac{P(M_i)P(x|M_i)}{\sum_{j=1}^2 P(M_j)P(x|M_j)}. \quad (2)$$

We have to compute the following quantities:

$$\begin{aligned} P(x|M_1) &= \int p_1(x|\lambda)\pi_1(\lambda)d\lambda = \int_0^{+\infty} \frac{e^{-5\lambda}\lambda^{23}}{4 \cdot 8! \cdot 10!} \cdot \frac{20^{100}\lambda^{100-1}e^{-20\lambda}}{\Gamma(100)} d\lambda \\ &= \frac{20^{100}\Gamma(123)}{25^{123} \cdot 4 \cdot 8! \cdot 10! \cdot \Gamma(100)} \int_0^{+\infty} \frac{e^{-25\lambda}\lambda^{123-1}25^{123}}{\Gamma(123)} d\lambda \\ &= \frac{20^{100}\Gamma(123)}{25^{123} \cdot 4 \cdot 8! \cdot 10! \cdot \Gamma(100)} \approx 2.5916 \cdot 10^{-7} \end{aligned}$$

Where we have forced to appear the probability density function of a random variable following a *Gamma*(123, 25) inside the integral, because the value of this integral is 1.

$$\begin{aligned} P(x|M_2) &= \int p_2(x|\theta)\pi_2(\theta)d\theta = \int_0^1 \theta^5(1-\theta)^{23} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1} d\theta \\ &= \frac{\Gamma(a+b)\Gamma(a+5)\Gamma(b+23)}{\Gamma(a)\Gamma(b)\Gamma(a+b+28)} \int_0^1 \frac{\Gamma(a+b+28)}{\Gamma(a+5)\Gamma(b+23)} \theta^{a+5-1}(1-\theta)^{b+23-1} d\theta \\ &= \frac{\Gamma(a+b)\Gamma(a+5)\Gamma(b+23)}{\Gamma(a)\Gamma(b)\Gamma(a+b+28)} \approx 1.9157 \cdot 10^{-6} \end{aligned}$$

Where a, b are the parameters computed before. Again, we have forced to appear the pdf of a random variable following in this case a *Beta*($a+5, b+23$) inside the integral, and then the integral is 1.

Now, we are able to compute the posterior probability for each model, and then decide which is the hypothesis we choose. Using the expression (2), we have that:

$$\begin{aligned} P(M_1|x) &= \frac{P(M_1)P(x|M_1)}{P(M_1)P(x|M_1) + P(M_2)P(x|M_2)} = \frac{0.5 \cdot 2.5916 \cdot 10^{-7}}{0.5 \cdot 2.5916 \cdot 10^{-7} + 0.5 \cdot 1.9157 \cdot 10^{-6}} \\ &\approx 0.119 \end{aligned}$$

Moreover,

$$P(M_2|x) = \frac{P(M_2)P(x|M_2)}{P(M_2)P(x|M_2) + P(M_1)P(x|M_1)} = \frac{0.5 \cdot 1.9157 \cdot 10^{-6}}{0.5 \cdot 1.9157 \cdot 10^{-6} + 0.5 \cdot 2.5916 \cdot 10^{-7}} \approx 0.881$$

And then,

$$FB_{21} = \frac{P(x|M_2)}{P(x|M_1)} = \frac{P(M_1)}{P(M_2)} \cdot \frac{P(M_2|x)}{P(M_1|x)} \approx 7.40 \quad (3)$$

Hence, we choose H_2 . It is, we choose that the data is generated with a Geometric model.

- Using the Bayesian model averaging approach:

Now, we are interested in using the Bayesian model averaging approach. Our goal is to plot the posterior predictive distribution, and once we have the posterior predictive distribution we can compute a point estimator and a credible interval. As we have seen in the theory lectures The model averaging consists of using a weighted average of the posterior predictive distributions of all models to make predictions. Thus, we will be using the following weighted average:

$$P(\tilde{x}|x) = P(M_1|x)P_1(\tilde{x}|x) + P(M_2|x)P_2(\tilde{x}|x). \quad (4)$$

As we can see in (4), we need first to compute the corresponding predictive posterior distribution for each model. In the case of the Poisson model, using the conjugation properties we know that the predictive posterior distribution follows a $NegBin(\alpha + \sum_{i=1}^n x_i, (\beta + n)/(1 + \beta + n)) = NegBin(123, 25/26)$. In the case of the Geometric model, despite the fact we have used a conjugate prior distribution, there is no closed form for the predictive posterior distribution. Then, what we are going to do is to use the simulation approach for compute the predictive posterior in the case of the Geometric model.

Let us first do a simulation for the predictive posterior in the case of the Poisson model. As we said, we know that the $P_1(\tilde{x}|x) \sim NegBin(123, 25/26)$, thus, we will simulate $M = 1000000$ values of a Negative Binomial with parameters 123 and 25/26. We have plotted these million of values and the result obtained is shown in the Figure 3.

Now, we want to compute the predictive posterior distribution of the second model, $P_2(\tilde{x}|x)$. In this case, we don't know the distribution analytically, so we will compute it by simulation. What we are going to do is simulate $M = 1000000$ values of a $Beta(a + 5, b + 23)$ (where the parameters a, b are the same than before), which is the posterior distribution of the Geometric model. And then, once we have these M values, we use them for simulate M values under

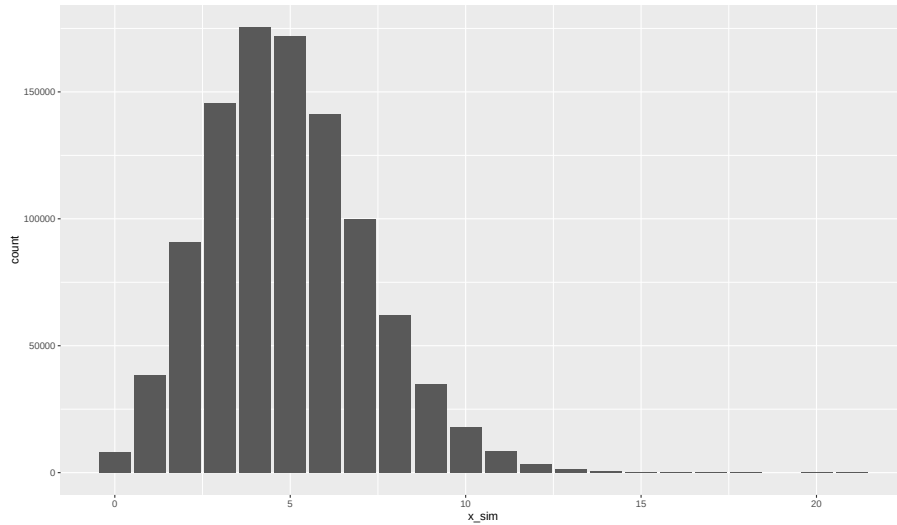


Figure 3: Bar graphic of the million of points simulated under $NegBin(123, 25/26)$.

the statistical model, which is the Geometric one. After develop this simulation, we have obtained the values represented in the bar graphic shown in Figure 4.

Now, we know how are the posterior predictive distributions of the two models. Our next goal is to use the model averaging approach and compute the posterior predictive of the models taking into account the weights given by $P(M_1|x) \approx 0.119$ and $P(M_2|x) \approx 0.881$. To do this, we take the M simulated values for the two predictive posteriors of the two models, having M pairs of simulated values (for each pair one value of the Poisson model and the other from the Geometric), and we choose only one value for each pair, we choose the value corresponding with the Poisson model with probability $P(M_1|x) \approx 0.119$, and the value corresponding to the Geometric model with probability $P(M_2|x) \approx 0.881$. Once we have done this, we have M simulated values corresponding to the averaged model, which is exactly what we want.

In practice, we have done the above procedure by simulating M values of a Bernoulli random variable with parameter 0.881, creating a data frame with the Bernoulli simulated values, the predictive posterior values from the Poisson model and the predictive posterior values from the Geometric model. And we have chosen the Geometric value if the Bernoulli simulated value is 1, and the Poisson value otherwise. By doing this we have obtained the values represented in the bar graphic exposed in the 5. As the reader can see, the shape is much more similar than the one observed in the Figure 4 than the one in the Figure 3. This is because the weight of the Geometric model is much bigger than the weight of the Poisson model.

Finally, once we have the values for the predictive posterior distribution for the model averaging computed, we can easily compute a point estimate, for instance the mean, and the

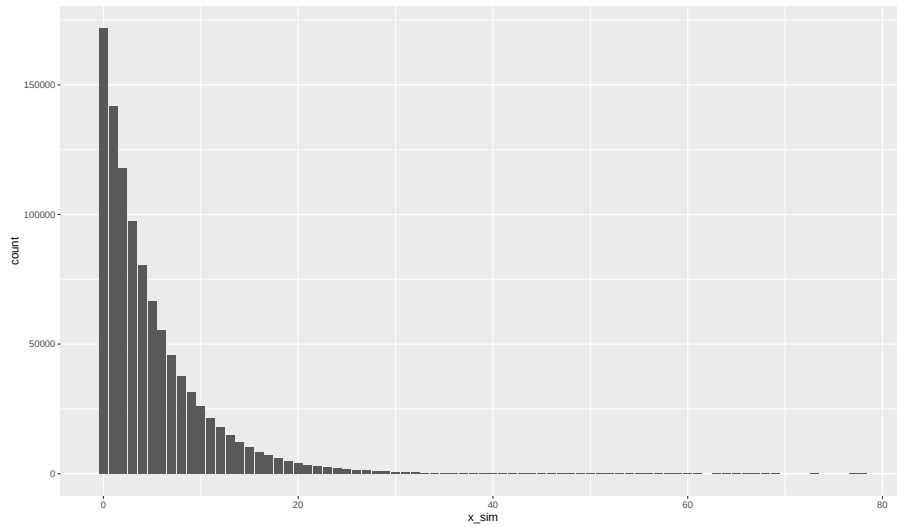


Figure 4: Bar graphic of the million of points of the posterior predictive distribution for the Geometric model.

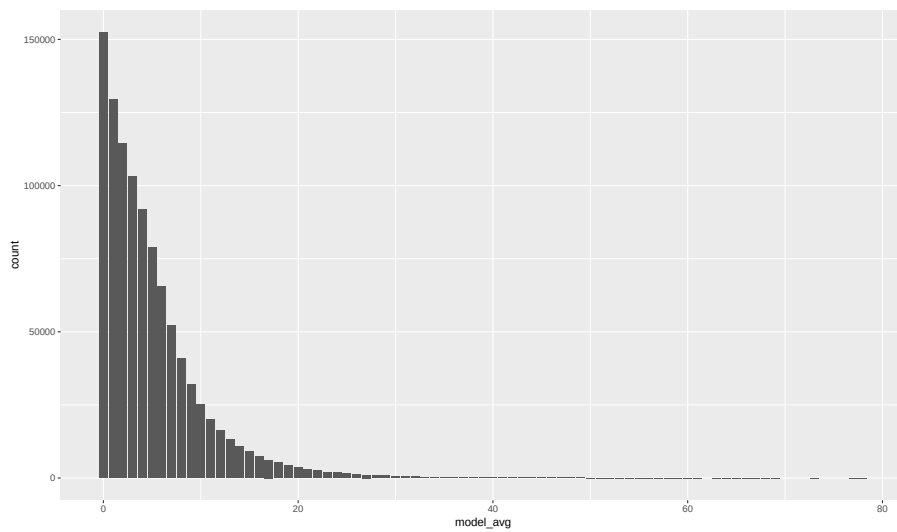


Figure 5: Bar graphic of the million of points of the posterior predictive distribution for the model averaging.

95% credible interval using R. We have obtained:

- A mean of 4.869253.
- The 95% credible interval: $CI_{95\%} = [0, 19]$.

The mean makes sense taking into account that our prior information was that the expected value of the counts is between 4 and 6 with a high probability. Regarding the 95% credible interval, we have that the count will be between 0 and 19 with a 95% of probability.

Homework optional

Arnau Garcia

2024-03-07

```
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(ggplot2)
```

Let us first draw in the same plot the prior, likelihood and posterior for the first model:

```
df2 <- tibble(
  lambda = seq(3, 7, 0.05),
  likelihood = ((95367431640625*exp(-5*lambda)*lambda^23)/41363226782215962624))

colors <- c("Prior" = "purple", "Likelihood" = "tomato", "Posterior" = "green")

plot1 <- ggplot() +
  geom_line(data=df2, aes(lambda, likelihood, color="Likelihood"))+
  labs(x="", y="")+
  stat_function(fun = dgamma,
    args = list(shape = 123, rate = 25),
    aes(color="Posterior")) +
  xlim(c(3, 7))+
  stat_function(fun = dgamma,
    args = list(shape = 100, rate = 20),
    aes(color="Prior")) +
  xlim(c(3, 7)) +
  labs(colour="")+
  scale_color_manual(values = colors) +
  theme_gray()
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```

Now we do the same but for the second model:

```
a <- 185306/1715
b <- 2686937/5145
df3 <- tibble(
```

```

theta = seq(0, 0.4, 0.0005),
likelihood = (2850120*theta^5*(1-theta)^(23)))

colors <- c("Prior" = "purple", "Likelihood" = "tomato", "Posterior" = "green")

plot2 <- ggplot() +
  geom_line(data = df3, aes(theta, likelihood, color = "Likelihood"))+
  labs(x="",y="")+
  stat_function(fun = dbeta,
               args = list(shape1 = a+5, shape2 = b+23),
               aes(color="Posterior")) +
  xlim(c(0, 0.4))+
  stat_function(fun = dbeta,
               args = list(shape1 = a, shape2 = b),
               aes(color="Prior")) +
  xlim(c(0, 0.4))+
  labs(colour="")+
  scale_color_manual(values = colors) +
  theme_gray()

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Let us simulate the predictive posterior distribution of the Poisson model, which is a negative binomial with parameters 123 and 25/26:

```

set.seed(999)
M<-1000000
y_pois <- rnbinoM(M, 123, 25/26)

plot3 <- ggplot(tibble(y_pois), aes(y_pois)) +
  labs(x="x_sim")+
  geom_bar()

```

Let us now simulate the predictive posterior for the Geometric model. We know that the posterior follows a Beta distribution with parameters $185306/1715 + 5$, $2686937/5145 + 23$.

```

#simulate M values form the posterior
theta_sim <- rbeta(M, shape1 = a+5, shape2 = b+23)
#simulate M values from the model using the above values
y_geom <- rgeom(M, theta_sim)

```

Let us now plot the previous values:

```

plot4 <- ggplot(tibble(y_geom), aes(y_geom)) +
  labs(x="x_sim")+
  geom_bar()

```

Now, we create a dataframe with three columns, the first one M realizations of a Bernoulli with success parameter 0.881, and the remaining two columns the simulations of the predictive posteriors done above:

```

bern <- rbinom(M,1,0.881)
df <- tibble(bern, y_pois, y_geom)
head(df)

```

A tibble: 6 x 3

bern y_pois y_geom

```
##      <int>  <int>  <int>
## 1         1      7      2
## 2         1      6      1
## 3         1      3      1
## 4         1     11      0
## 5         1      6      1
## 6         1      5      4
```

As we can see above, we have the previously described data frame. Now, we simulate the posterior predictive by means of the model averaging approach by taking the `y_geom` simulated value if `bern==1` and taking the `y_pois` otherwise.

```
model_avg <- ifelse(bern, y_geom, y_pois)
```

And now, we can plot the posterior predictive using the model averaging approach:

```
plot5 <- ggplot(tibble(model_avg), aes(model_avg)) +
  geom_bar()
```

Now, we can compute the point estimate and the 95% credible interval:

```
mean(model_avg)
```

```
## [1] 4.869253
```

```
quantile(model_avg, probs = c(0.025, 0.975))
```

```
## 2.5% 97.5%
```

```
##    0    19
```